Contest Problem Set 12120

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Name _____

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1. $2021 + 2021 =$ _____.
2. The tens digit of 961 is _____.
3. $60 \div 6 =$ _____.
4. $783 + 119 =$ _____.
5. $80 \times 15 =$ _____.
6. The remainder of $87 \div 7$ is _____.
7. $9 \times 6 - 9 =$ _____.
8. The product of 11 and 8 is _____.
9. $45 + 19 + 11 + 5 =$ _____.
10. (estimate) $199 + 399 + 796 + 597 =$ _____.
11. $453 - 363 =$ _____.
12. 1144 rounded to the nearest hundred is _____.
13. $125 \times 12 =$ _____.
14. $17^2 =$ _____.
15. $359 - 271 =$ _____.
16. $234 - 105 - 17 =$ _____.
17. DLXXVI in Arabic Numerals is _____.
18. $57 + 65 + 73 =$ _____.
19. $27 \times 13 =$ _____.
20. (estimate) $449 \times 401 =$ _____.
21. $40 \times 26 =$ _____.
22. The GCD of 18 and 30 is _____.
23. $399 \div 19 =$ _____.
24. If 1 yard is equal to 3 feet, then 81 feet is equal to _____ yards.
25. $32^2 =$ _____.
26. The larger of $\frac{8}{13}$ or $\frac{5}{8}$ is _____ (fraction).
27. $\frac{13}{24} + \frac{1}{12} =$ _____ (fraction).
28. $27^2 =$ _____.
29. Eight weeks is _____ days.
30. (estimate) $49 \times 51 \times 41 =$ _____.
31. $8 + 12 + 16 + 20 + 24 =$ _____.
32. The number of multiples of 7 between 29 and 64 is _____.
33. $37 \times 101 =$ _____.
34. $72 \times 78 =$ _____.
35. $3.091 \times 10^{-3} =$ _____ (decimal).
36. The sum of the prime numbers between 30 and 40 is _____.
37. The LCM of 18 and 30 is _____.
38. $54 \times 99 =$ _____.
39. The number of prime numbers that are between 30 and 40 is _____.
40. (estimate) $\sqrt{212121} =$ _____.

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41. $53^2 - 42^2 =$ _____.

42. The twelfth term in the arithmetic sequence $2, 5, 8, 11, 14, \ldots$ is _____.

43. $5^5 =$ _____.

44. The remainder of $472 \div 11$ is _____.

45. $111 \times 102 =$ _____.

46. The length of the diameter of a circle with an area of 64π is _____.

47. $\sqrt{1225} =$ _____.

48. The next term in the arithmetic sequence $2005, 2009, 2013, 2017, \ldots$ is _____.

49. $\frac{1}{4} + \frac{1}{12} + \frac{1}{36} =$ _____ (fraction).

50. (estimate) $142857 \times 16 =$ _____.

51. $\frac{2}{5} =$ _____ %.

52. $12\frac{1}{2}\% =$ _____ (fraction).

53. $78 \times 97 =$ _____.

54. $20^2 + 21^2 =$ _____.

55. The perimeter of an isosceles triangle with sides of length 18 and 8 is _____.

56. If $5x - 3 = 31$ then $10x - 3 =$ _____.

57. The sum of the terms in the arithmetic sequence $1, 3, 5, 7, \ldots, 17$ is _____.

58. 101101_2 in base 8 is _____ $_8$.

59. $\sqrt{7396} =$ _____.

60. (estimate) $161 \times 625 =$ _____.

61. The number 101 written in base 7 is _____$_7$.

62. $10101 \div 13 =$ _____.

63. A fair coin is flipped four times. The probability of exactly three heads is _____ (fraction).

64. $7\frac{4}{5} \times 7\frac{1}{5} =$ _____ (mixed number).

65. If $4^{x+2} = 250$ then $4^{x+3} =$ _____.

66. The number 40 written in base 6 is _____$_6$.

67. Two angles in a triangle have measures 59° and 42°, and the third angle has a measure of ___ $^\circ$.

68. $0.\overline{42} =$ _____;(fraction).

69. $10\frac{5}{8} \times 14\frac{5}{8} =$ _____ (mixed number).

70. (estimate) $(\sqrt{5})^7 =$ _____.

71. $\sqrt[5]{161051} =$ _____.

72. The number of positive whole number divisors of 70 is _____.

73. The hypotenuse of a right triangle with legs of length 9 and 40 is _____.

74. The reciprocal of $\frac{8}{31}$ is _____ (decimal).

75. An equilateral triangle with height $4\sqrt{3}$ has a side of length _____.

76. $52 \times 60 \times 15 =$ _____.

77. The mean value of the set $\{41, 43, -16, 42, 40\}$ is _____.

78. The tenth term in the geometric sequence $1, 2, 4, 8, \ldots$ is _____.

79. The area of a right triangle with a hypotenuse of length 17 and one leg of length 8 is _____.

80. (estimate) $\sqrt{5} \cdot \sqrt{101} \cdot \sqrt{8} \cdot \sqrt{2} =$ _____.

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Select E only if you cannot determine a uniquely correct answer between A, B, C, and D.

1.	A	B	C	D	E	11.	A	B	C	D	E	21.	A	B	C	D	E
2.	A	B	C	D	E	12.	A	B	C	D	E	22.	A	B	C	D	E
3.	A	B	C	D	E	13.	A	B	C	D	E	23.	A	B	C	D	E
4.	A	B	C	D	E	14.	A	B	C	D	E	24.	A	B	C	D	E
5.	A	B	C	D	E	15.	A	B	C	D	E	25.	A	B	C	D	E
6.	A	B	C	D	E	16.	A	B	C	D	E	26.	A	B	C	D	E
7.	A	B	C	D	E	17.	A	B	C	D	E	27.	A	B	C	D	E
8.	A	B	C	D	E	18.	A	B	C	D	E	28.	A	B	C	D	E
9.	A	B	C	D	E	19.	A	B	C	D	E	29.	A	B	C	D	E
10.	A	B	C	D	E	20.	A	B	C	D	E	30.	A	B	C	D	E

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1. A bag of potato chips costs \$1.95. A bottle of orange juice costs \$4.95. How much do two bottles of orange juice and one bag of potato chips cost?

(A) \$8.85 (B) \$11.65 (C) \$11.85 (D) \$8.65 (E) Other

2. Mark drew one shape on each sheet of fifteen sheets of paper. On four of the sheets, he drew triangles. On eight of the sheets, he drew rectangles. On three of the sheets, he drew pentagons. How many total sides did Mark draw?

(A) 52 (B) 59 (C) 64 (D) 50 (E) Other

3. Sahaj picked a number. He then doubled it, subtracted twenty, and finally tripled that result, producing the number 24. What was Sahaj's original number?

(A) 14 (B) 6 (C) 10 (D) 12 (E) Other

4. A fortnight is two weeks. A week is seven days. If Peter's trip was six fortnights long, how long was Peter's trip in days?

(A) 70 (B) 98 (C) 84 (D) 91 (E) Other

5. What is the value of $11 \times 9 \times 13 \times 15$?

(A) 19295 (B) 19405 (C) 19495 (D) 19305 (E) Other

6. A nickel is worth \$0.05, and a dime is worth \$0.10. Emily has three dozen nickels and two dozen dimes in her purse. How much money, in dollars, does Emily have in her purse?

(A) \$4.80 (B) \$4.20 (C) \$3.60 (D) \$4.50 (E) Other

7. If fifteen los are equal to nine las, then how many los are equal to ninety-nine las?

(A) 135 (B) 105 (C) 165 (D) 81 (E) Other

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8. The latest movie starring superspy agent Ethan was 147 minutes long. If a showing of that movie started at 6 : 15 PM, then at what time did the showing end?

(A) 8 : 42 PM (B) 9 : 02 PM (C) 8 : 22 PM (D) 8 : 02 PM (E) Other

9. What is the tens digit of 2021×2020?

(A) 4 (B) 0 (C) 8 (D) 6 (E) Other

10. Nicole has a collection of four cubes, three triangular prisms, and one square pyramid. What is the total number of faces in all of the 3D objects of Nicole's collection?

(A) 41 (B) 44 (C) 43 (D) 32 (E) Other

11. What is the value of $20 - 19 + 22 - 21 + 24 - 23 + 26 - 25 + 28 - 27 + 30 - 29$?

(A) 6 (B) 5 (C) -6 (D) -5 (E) Other

12. How many twelves are in sixteen nines?

(A) 16 (B) 12 (C) 24 (D) 32 (E) Other

13. A bag contains a red marble, a yellow marble, a green marble, and a blue marble. MacKenzie reaches into the bag, removes a marble, and sets it aside. She then does the same thing two more times. What is the probability that the marble remaining in the bag is red?

(A) $\frac{3}{4}$ (B) $\frac{3}{8}$ (C) $\frac{1}{4}$ (D) $\frac{1}{2}$ (E) Other

14. In the year 2020, October has 31 days, November has 30 days, and December has 31 days. To the nearest whole number, how many complete weeks are in the last three months of 2020?

(A) 13 (B) 15 (C) 14 (D) 12 (E) Other

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15. A rectangle with sides of length 84 and 45 has the same perimeter as a hexagon. If the sides of the hexagon all have the same length, then what is the length of a side of the hexagon?

(A) 41 (B) 46 (C) 43 (D) 45 (E) Other

16. Emma's cake recipe requires three large eggs to make two dozen slices of cake. With the same recipe, how many large eggs are needed to make 144 slices of cake?

(A) 16 (B) 8 (C) 9 (D) 12 (E) Other

17. What is the sum of the six positive whole numbers that evenly divide 20?

(A) 30 (B) 33 (C) 45 (D) 42 (E) Other

18. A square has an area of 25. Two opposite sides of the square each have their length tripled. What is the perimeter of the new figure?

(A) 40 (B) 25 (C) 60 (D) 35 (E) Other

19. Becca's birthday is June 10th. Her friend's birthday is November 10th. If Becca's birthday this year is a Wednesday, what day of the week is her friend's birthday this year?

(A) Wednesday (B) Friday (C) Saturday (D) Monday (E) Other

20. A circle has an area of 25π. The radius of the circle is increased by 3. What is the increase in the area of the circle?

(A) 24π (B) 64π (C) 39π (D) 36π (E) Other

21. In a modified ladder toss game, all scores are worth either 4 points or 9 points. Hunter made 100 scores when playing that game, and his point total was 755 points. How many more 9 point scores than 4 point scores did Hunter made?

(A) 42 (B) 44 (C) 41 (D) 43 (E) Other

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22. Thomas lists all of the different arrangements of the letters in the word *CABOOSE*. How many of Thomas's arrangements begin with a *C* and end with an *E*?

(A) 100 (B) 120 (C) 80 (D) 90 (E) Other

23. If $a\clubsuit b = ab - a - b + 1$, then what is the value of $284\clubsuit 1001$?

(A) 282435 (B) 283000 (C) 283283 (D) 282000 (E) Other

24. What is the product of 4% of 112 and $6\frac{1}{4}$% of 375?

(A) 144 (B) 90 (C) 120 (D) 105 (E) Other

25. Josh rolls two standard six-sided dice. What is the probability that the product of the numbers shown on the dice is a multiple of six?

(A) $\frac{4}{9}$ (B) $\frac{1}{3}$ (C) $\frac{5}{12}$ (D) $\frac{1}{2}$ (E) Other

26. What is the remainder when the sum of the terms of the arithmetic series $101 + 108 + 115 + \cdots + 290$ is divided by 28?

(A) 7 (B) 0 (C) 14 (D) 21 (E) Other

27. Two right triangles have the same perimeter, and each triangle's side lengths are all whole numbers. One right triangle has legs of length 10 and 24. The other right triangle has a hypotenuse of length 25. What is the area of the second triangle?

(A) 150 (B) 112 (C) 120 (D) 144 (E) Other

28. A function f is defined for positive integers x as $f(x) = 0$ if $x \leq 2$. Otherwise, $f(x) = f(x-1)$ if x is not divisible by 3, and $f(x) = 2 + f\left(\frac{x}{3}\right)$ if x is divisible by 3. What is the value of $f(2020)$?

(A) 16 (B) 14 (C) 12 (D) 8 (E) Other

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29. A rectangle measures 6 centimeters by 25 centimeters. The probability that a point randomly chosen inside the rectangle is closer to a side of length 25 than a side of length 6 is $\frac{p}{q}$, where p and q are relatively prime positive whole numbers. What is the sum of p and q?

(A) 5 (B) 19 (C) 37 (D) 47 (E) Other

30. The three values of x that are solutions to $x^3 + mx = 37x^2 + n$ are positive whole numbers that form a geometric sequence. What is the sum of the possible values of the common ratio of the geometric sequence?

(A) $\frac{25}{12}$ (B) $\frac{9}{4}$ (C) $\frac{7}{4}$ (D) 2 (E) Other

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Problems 1 & 2

1. Mabel and Ann each picked a positive whole number. The sum of Mabel's and Ann's numbers was 95. The product of Mabel's and Ann's numbers was 364. If Mabel's number was greater than Ann's number, what is the value of Mabel's number minus Ann's number?

1.

2. Jada has two $10 bills and three $1 bills. Chase has the same total amount of money as Jada. Chase has three $5 bills, and the rest of his money consists of coins worth $0.05 each. How many $0.05 coins does Chase have?

2.

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Problems 3 & 4

3. A four-person math relay contest involves each person writing a numerical answer and then passing the result to another teammate. Diana wrote an answer, and Emma received Diana's answer. Emma's answer was four times Diana's answer, and Kassy received Emma's answer. Kassy's answer is 25 more than Emma's answer, and Michaela received Kassy's answer. Michaela's answer was 247, which was one third of Kassy's answer. What was Diana's answer?

3.

4. A spool of yarn is 24 yards long. There are 3 feet in one yard and 12 inches in one foot. How many pieces of yarn that are exactly 16 inches long can Phillip produce from 3 of these spools of yarn?

4.

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Problems 5 & 6

5. What is the sum of all the digits of $9 \times 8 \times 7 \times 6 \times 4 \times 3 \times 2 \times 1$?

5.

6. A rectangle has whole number side lengths and an area of 126. What is the sum of all possible different perimeters of the rectangle?

6.

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Problems 7 & 8

7. Aidan received 36 action figures on the day of his 18th birthday. He gave 25% of the action figures to Brandon. Aidan then gave 2 action figures to Jack. Finally, Aidan gave 40% of the remaining action figures to Chris. Aidan kept the rest of the action figures for himself. How many action figures did Aidan keep for himself?

7.

8. How many three-digit positive whole numbers have digits that sum to no more than 10?

8.

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Team Round
12120

School or Team

Name _____

Name _____

Name _____

Name _____

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1.	2.	3.	4.	5.
6.	7.	8.	9.	10.

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1. Some of the coins in Mexico represent a different number of centavos. Marian, Emma, and Tanya each have collections of coins worth 500 centavos. Marian's coins are all 10 centavo coins, Emma's coins are all 20 centavo coins, and Tanya's coins are all 50 centavo coins. How many total coins do Marian, Emma, and Tanya have altogether?

2. Tyler sells painted kendamas at the school fair. He can purchase one dozen unpainted kendamas at a total cost of $10 and can sell each painted kendama for $2. Tyler buys fifteen dozen unpainted kendamas. How much profit, in dollars, does Tyler make if he paints and sells all of his kendamas at the school fair? Profit is total money received minus total money spent.

3. A regular pentakaidecagon is a shape with 15 sides that are all equal in length. A regular tetracontagon is a shape with 40 sides that are all equal in length. A regular pentakaidecagon and a regular tetracontagon each have perimeters 960. How much greater is the side length of the pentakaidecagon than the side length of the tetracontagon?

4. What is the sum of the first ten odd positive composite whole numbers? Composite numbers are whole numbers greater than 2 that are not prime.

5. In a neighborhood parade, 25 children each rode either a bicycle or a tricycle. A bicycle has two wheels and a tricycle has three wheels. Altogether, the number of wheels on the vehicles ridden by the children totaled 53. How many more children rode bicycles than rode tricycles?

6. What is the sum of all of the positive whole numbers less than 120 that are multiples of 3 but are not multiples of 6?

7. A large dance hall has a dance floor that measures 58 feet by 98 feet. The owner of the dance hall covers the dance floor in tiles that are squares measuring 2 feet by 2 feet. The tiles are either black or white, and the owner covers the floor in a checkerboard pattern. One of the tiles in a corner of the dance floor is white. When tiling the floor, how many black tiles will the dance hall owner need?

8. Nick rolls two standard six-sided dice and then multiplies the numbers shown. What is the probability that Nick's result is less than 20? Express your answer as a common fraction.

9. John was born on April 23, 1977. On April 23, 2021, how many weeks old is John? Express your answer to the nearest whole number. Note that 2000 is a leap year.

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10. When written in binary, or base 2, how many positive whole numbers have 6 or fewer digits and 3 or fewer 0's?

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Answers 12120

Sprint Round

1. C
2. B
3. A
4. C
5. D
6. B
7. C
8. A
9. E
10. B
11. A
12. B
13. C
14. A
15. C
16. E
17. D
18. A
19. E
20. C
21. A
22. E
23. B
24. D
25. C
26. C
27. A
28. C
29. D
30. A

Target Round

1. 87
2. 160
3. 179
4. 162
5. 27
6. 624
7. 15
8. 219

Team Round

1. 85
2. 210
3. 40
4. 298
5. 19
6. 1200
7. 710
8. $\frac{7}{9}$
9. 2296
10. 56

Number Sense

1. 4042
2. 6
3. 10
4. 902
5. 1200
6. 3
7. 45
8. 88
9. 80
10. $[1892, 2090]$
11. 90
12. 1100
13. 1500
14. 289
15. 88
16. 112
17. 576
18. 195
19. 351
20. $[171047, 189051]$
21. 1040
22. 6
23. 21
24. 27
25. 1024
26. $\frac{5}{8}$
27. $\frac{5}{8}$
28. 729
29. 56
30. $[97337, 107581]$
31. 80
32. 5
33. 3737
34. 5616
35. 0.003091
36. 68
37. 90
38. 5346
39. 2
40. $[438, 483]$
41. 1045
42. 35
43. 3125
44. 10
45. 11322
46. 16
47. 35
48. 2021
49. $\frac{13}{36}$
50. $[2171427, 2399997]$
51. 40
52. $\frac{1}{8}$
53. 7566
54. 841
55. 44
56. 65
57. 81
58. 55
59. 86
60. $[95594, 105656]$
61. 203
62. 777
63. $\frac{1}{4}$
64. $56\frac{4}{25}$
65. 1000
66. 104
67. 79
68. $\frac{14}{33}$
69. $155\frac{25}{64}$
70. $[266, 293]$
71. 11
72. 8
73. 41
74. 3.875
75. 8
76. 46800
77. 30
78. 512
79. 60
80. $[86, 94]$

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Solutions 12120

Sprint Round Solutions

1. Two bottles of orange juice cost $\$4.95 + \$4.95 = \$9.90$, and adding a bag of potato chips brings the total to $\$9.90 + \$1.95 = \boxed{\$11.85}$.

2. The 4 sheets with triangles totaled $4 \cdot 3 = 12$ sides. The 8 sheets with rectangles totaled $8 \cdot 4 = 32$ sides. The 3 sheets with pentagons totaled $3 \cdot 5 = 15$ sides. Altogether, the number of sides was $12 + 32 + 15 = \boxed{59}$.

3. Before tripling, the result was $24 \div 3 = 8$. Before subtracting 20, the result was $8 + 20 = 28$. And before doubling, the original number was $28 \div 2 = \boxed{14}$.

4. Six fortnights is $6 \cdot 2 = 12$ weeks, and the number of days in Peter's 12-week trip is $7 \cdot 12 = \boxed{84}$.

5. We have $9 \times 11 = 99$ and $13 \times 15 = 195$, and 99×195 is $195 \times 100 - 195$, or $19500 - 195 = \boxed{19305}$.

6. The value of the nickels is $3 \cdot 12 \cdot 0.05 = 1.80$ dollars. The value of the dimes is $2 \cdot 12 \cdot 0.10 = 2.40$ dollars. Emily has in total $\$1.80 + \$2.40 = \boxed{\$4.20}$.

7. A quantity of 99 las is $\frac{99}{9} = 11$ times the number in the known exchange. The number of los in the known exchange is 15, and multiplying this by the same factor of 11 yields $11 \cdot 15 = \boxed{165}$.

8. There are 60 minutes in an hour, and $147 = 2 \cdot 60 + 27$, so the length of the movie is 2 hours and 27 minutes. A time 2 hours later than $6:15$ is $8:15$, and a time 27 minutes after $8:15$ is $\boxed{8:42\text{ PM}}$.

9. As 2020 is a multiple of 10, the units digit of the product is 0, and the tens digit will be the units digit of 2021×202. This is the units digit of 1×2, which is $\boxed{2}$.

10. Each of the 4 cubes has 6 faces, each of the 3 triangular prisms has 5 faces, and the square pyramid has 5 faces. Altogether, the number of faces is $4 \times 6 + 3 \times 5 + 5 = \boxed{44}$.

11. Group the expression as $(20 - 19) + (22 - 21) + (24 - 23) + (26 - 25) + (28 - 27) + (30 - 29)$, which is $1 + 1 + 1 + 1 + 1 + 1 = \boxed{6}$.

12. Sixteen nines is $16 \times 9 = 144$, and the number of twelves is $144 \div 12 = \boxed{12}$.

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13. We could do casework on the draws, but note that there is an equal probability of any 1 of the 4 marbles being the one remaining in the bag. No marble is more likely than any other to be the one left in the bag. Therefore, the probability is $\boxed{\frac{1}{4}}$.

14. The total number of days in the last three months of 2020 is $31 + 30 + 31 = 92$, so the number of complete weeks is $92 \div 7 \approx 13.143$, which rounded to the nearest whole number is $\boxed{13}$.

15. The perimeter of the rectangle is $2 \cdot (84+45) = 258$, so the length of a side of the hexagon is $\frac{258}{6} = \boxed{43}$.

16. Two dozen slices is $2 \cdot 12 = 24$ slices. Then 1 large egg is needed for $24 \div 3 = 8$ slices of cake. The number of large eggs required to make 144 slices is $144 \div 8 = \boxed{18}$.

17. The 6 positive whole numbers that evenly divide 20 are 1, 2, 4, 5, 10, and 20. Their sum is $1+2+4+5+10+20 = \boxed{42}$.

18. Each side of the square has a length of $\sqrt{25} = 5$. The new figure has two sides of length $3 \cdot 5 = 15$ and two sides of length 5, for a perimeter of $15 + 15 + 5 + 5 = \boxed{40}$.

19. There are $30 - 10 = 20$ remaining days in June, plus 31 days in July, plus 31 days in August, plus 30 days in September, plus 31 days in October, plus 10 days for a total of $20+31+31+30+31+10 = 153$ days from June 10th to November 10th. Since $153 \div 7$ has a remainder of 6, November 10th will be 1 day earlier in the week than Wednesday, which is $\boxed{\text{Tuesday}}$.

20. Since the area of a circle is $\pi \cdot r^2$, the radius of a circle with an area of 25π is $\sqrt{25} = 5$. Increasing this radius by 3 gives a new radius of $5 + 3 = 8$ and a new area of $\pi \cdot 8^2 = 64\pi$, for an increase in area of $64\pi - 25\pi = \boxed{39\pi}$.

21. Let F be the number of four point scores and N be the number of nine point scores. Then $F + N = 100$ and $4F + 9N = 755$. Multiplying the first equation by -4 and adding to the second yields $5N = 355$, so $E = 71$. Therefore F is $100 - 71 = 29$, and the number of 9 point scores exceeds the number of 4 point scores by $71 - 29 = \boxed{42}$.

22. Once the C is placed in the first position and the E is placed in the last position, we only need to count the number of different ways to arrange the letters $ABOOS$. There are 5 letters, so they can be arranged in $5! = 120$ ways. But we must also account for the two O's being indistinguishable, which divides the result by $2! = 2$, Thus the total number of different arrangements is $\frac{120}{2} = \boxed{60}$.

23. This could be calculated directly, but note that $ab - a - b + 1 = (a-1)(b-1)$. Therefore, the value is $(284-1)(1001-1) = 283 \cdot 1000$, or $\boxed{283000}$.

24. Observing that 4% is $\frac{1}{25}$ and $6\frac{1}{4}$% is $\frac{1}{16}$, the product is $\frac{1}{25} \cdot 112 \cdot \frac{1}{16} \cdot 375$, or $\frac{1}{25} \cdot 375 \cdot \frac{1}{16} \cdot 112$. This is $15 \cdot 7 = \boxed{105}$.

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25. There are $6 \cdot 6 = 36$ outcomes. The product will be a multiple of 6 if either die shows a 6 or if one die is a 2 or a 4 and the other is a 3. By drawing a table of possibilities, we find that there are 15 outcomes where the product is a multiple of 6, for a probability of $\frac{15}{36} = \boxed{\frac{5}{12}}$.

26. The series contains $\frac{290-101}{7} + 1 = 28$ terms, with an average value of $\frac{101+290}{2} = 195\frac{1}{2}$. Therefore, the sum is $28 \cdot 195\frac{1}{2}$, or $28 \cdot 195 + 14$. The remainder when this product is divided by 28 is $\boxed{14}$.

27. By the Pythagorean theorem, the hypotenuse of the first right triangle is $\sqrt{10^2 + 24^2} = 26$, so its perimeter is $10 + 24 + 26 = 60$. Therefore the two legs of the second triangle must have lengths that sum to $60 - 25 = 35$, and their squares of their lengths must sum to $25^2 = 625$. Trying different squares, we see that $15 + 20 = 35$ and $15^2 + 20^2 = 625$, so the area of the second triangle is $\frac{1}{2} \cdot 15 \cdot 20 = \boxed{150}$.

28. Applying the definition of the function, $f(2020) = f(2019) = 2 + f(673)$, and $f(673) = f(672)$. Then $f(2020) = 4 + f(224)$, and $f(224) = f(222)$. Then $f(2020) = 6 + f(74)$, and $f(74) = f(72)$. Then $f(2020) = 8 + f(24)$, and $f(2020) = 10 + f(8)$. Since $f(8) = f(6)$, $f(2020) = 12 + f(2)$, and $f(2) = 0$. Thus $f(2020) = \boxed{12}$.

29. The point will only be closer to the side of length 25 if it lies outside of two congruent isosceles right triangles with hypotenuses on the sides of length 6 (note that the two right triangles will not overlap). The area of the entire rectangle is $6 \cdot 25 = 150$. The area of the isosceles right triangles is $2 \cdot \left(\frac{1}{4} \cdot 6^2\right) = 18$. The area of the region outside these triangles is $150 - 18 = 132$, for a probability of $\frac{132}{150} = \frac{22}{25}$, and $22 + 25 = \boxed{47}$.

30. Rewrite the equation as $x^3 - 37x^2 + mx - n = 0$. Let the positive whole numbers be a, ar, and ar^2. By Vieta's formulas, the sum of the roots of the cubic is 37, so $a + ar + ar^2 = 37$, or $a \cdot (1 + r + r^2) = 37$. Since a is a whole number but 37 is prime, r must be a fraction of the form $\frac{p}{q}$, where p and q are positive whole numbers. Then the central term can be written as pq, the first term as q^2, and the third term as p^2. We can now try pairs of squares with sums less than 37, and see that $p^2 = 9$, $pq = 12$, and $q^2 = 16$ are solutions since $9 + 12 + 16 = 37$. So the two possible values of the common ratio of the geometic sequence are $\frac{12}{9} = \frac{4}{3}$ and $\frac{9}{12} = \frac{3}{4}$, and their sum is $\frac{4}{3} + \frac{3}{4} = \boxed{\frac{25}{12}}$.

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Target Round Solutions

1. Factor pairs of 364 in least to greatest order are $(1, 364)$, $(2, 182)$, $(4, 91)$, $(7, 52)$, $(13, 28)$, and $(14, 26)$. The pair that sums to 95 is $(4, 91)$, and $91 - 4 = \boxed{87}$.

2. Jada has a total of $2 \cdot \$10 + 3 \cdot \$1 = \$23$. Chase has $3 \cdot \$5 = \$15$ in \$5 bills, so he has $\$23 - \$15 = \$8$ in \$0.05 coins. There are $\frac{1}{0.05} = 20$ such coins in \$1, so Chase's total number of coins is $20 \cdot 8 = \boxed{160}$.

3. Working backwards, since Michaela's result was 247, Kassy's result was $247 \cdot 3 = 741$. Then Emma's result was $741 - 25 = 716$, and Diana's number was $716 \div 4 = \boxed{179}$.

4. Each spool of yarn in $24 \cdot 3 = 72$ feet long, which is $72 \cdot 12 = 864$ inches. The number of 16-inch pieces from one of these spools is $\frac{864}{16} = 54$. The number of pieces from 3 spools is $3 \cdot 54 = \boxed{162}$.

5. This is not simply 9!, as the factor of 5 is not included, but the product can be calculated as $9! \div 5 = 72576$. The sum of all the digits of the result $7 + 2 + 5 + 7 + 6 = \boxed{27}$.

6. The length and width of the rectangle must be a factor pair of 126. The possible pairs in least to greatest order are $(1, 126)$, $(2, 63)$, $(3, 42)$, $(6, 21)$, $(7, 18)$, and $(9, 14)$. The perimeter of the rectangle is twice the sum of the length and width, and from the Distributive Property, the sum of all possible perimeters is $2 \cdot (1 + 126 + 2 + 63 + 3 + 42 + 6 + 21 + 7 + 18 + 9 + 14) = \boxed{624}$.

7. Aidan started with 36 action figures. He gave 25% to Brandon, so Aidan kept 75%, or $36 \cdot \frac{3}{4} = 27$ action figures. After giving 2 action figures to Jack, he had $27 - 2 = 25$ action figures left. He gave 40% to Chris, so he kept 60% of the 25 action figures, and $25 \cdot 0.6 = \boxed{15}$.

8. We can approach the problem using "sticks and stones". We are distributing no more than 10 identical items into three containers (each representing a different digit). If there are leftovers, we can put them in a fourth container (making the sum always 10). For example, for the three-digit number 123, the fourth container would have $10 - (1 + 2 + 3) = 4$ items. The hundreds digit must be at least 1, so we are then distributing 9 remaining items among 4 different containers. By sticks and stones, this is $\binom{12}{9} = 220$. But we must discount the case where all 10 items end up in the container for the hundreds digit (since there is no digit greater than 9). Thus, the total number of such three-digit numbers is $220 - 1 = \boxed{219}$.

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Team Round Solutions

1. Marian has $\frac{500}{10} = 50$ coins that are each worth 10 centavos. Emma has $\frac{500}{20} = 25$ coins that are each worth 20 centavos. Tanya has $\frac{500}{50} = 10$ coins that are each worth 50 centavos. Altogether, their total number of coins is $50 + 25 + 10 = \boxed{85}$.

2. The total cost of 15 dozen kendamas is $15 \cdot \$10 = \$150$. The total revenue (money received) from selling $15 \cdot 12 = 180$ kendamas is $\$180 \cdot 2 = \$360$. The profit is $\$360 - \$150 = \boxed{\$210}$.

3. The side length of the pentakaidecagon is $\frac{960}{15} = 64$. The side length of the tetracontagon is $\frac{960}{40} = 24$. The side length of the pentakaidecagon exceeds the side length of the tetracontagon by $64 - 24 = \boxed{40}$.

4. By going through the first few odd numbers, the first ten odd composite numbers are 9, 15, 21, 25, 27, 33, 35, 39, 45, and 49. The sum of those numbers is $\boxed{298}$.

5. If every child rode a bicycle, then there would be $25 \cdot 2 = 50$ wheels. Each bicycle that is replaced by a tricycle adds 1 wheel, so the number of tricycles is $53 - 50 = 3$. Therefore, the number of bicycles is $25 - 3 = 22$, and the number of bicycles exceeds the number of tricycles by $22 - 3 = \boxed{19}$.

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6. The positive whole numbers less than 120 that are multiples of 3 but are not multiples of 6 are 3, 9, 15, 21, 27, 33, 39, 45, 51, 57, 63, 69, 75, 81, 87, 93, 99, 105, 111, and 117. Now note that $3 + 117 = 120$, $9 + 111 = 120$, and so on. There are a total of $\frac{20}{2} = 10$ pairs of numbers that sum to 120, so the sum of the numbers is $120 \cdot 10 = \boxed{1200}$.

7. The owner needs $\frac{58}{2} = 29$ columns and $\frac{98}{2} = 49$ rows of tiles. One corner tile is white, so since the number of rows and columns are both odd, all corner tiles will be white. Therefore, there will be 25 rows with 14 black tiles, and 24 rows with 15 black tiles. The total number of black tiles is $25 \cdot 14 + 24 \cdot 15 = \boxed{710}$.

8. As each die has 6 outcomes, there are $6 \cdot 6 = 36$ possible outcomes for the product. Of these, the only outcomes with products greater than or equal to 20 are $(4, 5)$, $(4, 6)$, $(5, 4)$, $(5, 5)$, $(5, 6)$, $(6, 4)$, $(6, 5)$, and $(6, 6)$, for a total of 8. Therefore the number of possible outcomes with a product less than 20 is $36 - 8 = 28$. The probability the product is less than 20 is $\frac{28}{36} = \boxed{\frac{7}{9}}$.

9. Note that there are $2021 - 1977 = 44$ years between the two given dates, and 11 of the years are leap years. Each year is 365 days and each leap year is 366 days. Since a week is seven days, there are a total of $\frac{365 \cdot 33 + 366 \cdot 11}{7} = \frac{12045 + 4026}{7} = \frac{16071}{7} \approx 2295.857$ weeks. To the nearest whole number, this is $\boxed{2296}$.

10. Any binary number with 4 or fewer digits will have 3 or fewer 0's, and there are $2^4 - 1 = 15$ such numbers. For binary numbers with 5 digits, the leftmost digit must be a 1, and any 0, 1, or 2 of the remaining 4 digits may be 0. There are $\binom{4}{0} + \binom{4}{1} + \binom{4}{2} + \binom{4}{3} = 15$ such numbers. For binary numbers with 6 digits, the leftmost digit must be a 1, and any 0, 1, or 2 of the remaining 5 digits may be 0. There are $\binom{5}{0} + \binom{5}{1} + \binom{5}{2} + \binom{5}{3} = 26$ such numbers. Altogether, the total number of such numbers is $15 + 15 + 26 = \boxed{56}$.